

PROJECT SUMMARY

Project: Northgate Resort Tower C

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Executive Project Summary: Denver 3-Story Office Building

STRUCTMIND AI Report | Date: 23-MAY-2024 | Project ID: DNV-S3-180T

Project Description

This project comprises the complete structural steel package for a new three-story, approximately 30,000 sqft commercial office building in Denver, Colorado. The structural system utilizes a conventional steel frame with moment frames for lateral stability and composite metal decking for floor systems. The scope includes the detailing, fabrication, delivery, and erection of all primary and secondary structural steel members. The design is intended to provide flexible Class-A office space and support a modern building envelope.

Key Quantities

- Total Tonnage: 180 tons
- Total Pieces: Approx. 2,100 members (columns, beams, braces)
- Total Connections: Approx. 4,000 bolted and welded connections

Top 5 Risks

1. Steel Mill Lead Times & Price Volatility: Extended mill rolling cycles and fluctuating raw material costs pose a significant threat to budget and schedule adherence.
2. Skilled Labor Shortages: The availability of certified ironworkers and fabricators in the Denver market could impact erection speed and fabrication quality.
3. Constrained Site Logistics: A tight urban footprint may limit material laydown areas and crane access, requiring meticulous just-in-time delivery coordination.
4. Adverse Weather Delays: Denver's potential for high winds and winter snow accumulation presents a direct risk to the erection schedule and site safety.
5. MEP/Façade Coordination: Incomplete coordination between structural steel shop drawings and MEP/façade models could lead to costly field modifications and RFIs.

Top 5 Opportunities

1. **Early BIM Integration & Automated Fabrication:** Leveraging a fully integrated BIM model from design through fabrication can drastically reduce errors and accelerate shop drawing approval.
2. **Phased Erection & Just-in-Time Delivery:** A well-planned erection sequence can allow other trades (e.g., deckers, concretors) to begin work on lower floors while upper floors are still being erected, compressing the overall project schedule.
3. **Strategic Sourcing of Domestic Steel:** Prioritizing domestic mills with established delivery records can mitigate international supply chain risks and potentially qualify for project incentives.
4. **Off-site Prefabrication of Assemblies:** Prefabricating complex assemblies (e.g., stair towers, braced-frame modules) on the ground or off-site can improve safety, quality control, and reduce crane time.
5. **Value Engineering for Connection Optimization:** A focused review of connection designs could identify opportunities to substitute bolted for welded connections, reducing field labor hours and inspection requirements.

Recommended Schedule Horizon

A total duration of 20-24 weeks is recommended from contract award to structural steel topping out.

- Detailing & Approval: 6 weeks
- Material Procurement & Fabrication: 12 weeks
- Site Erection: 4 weeks